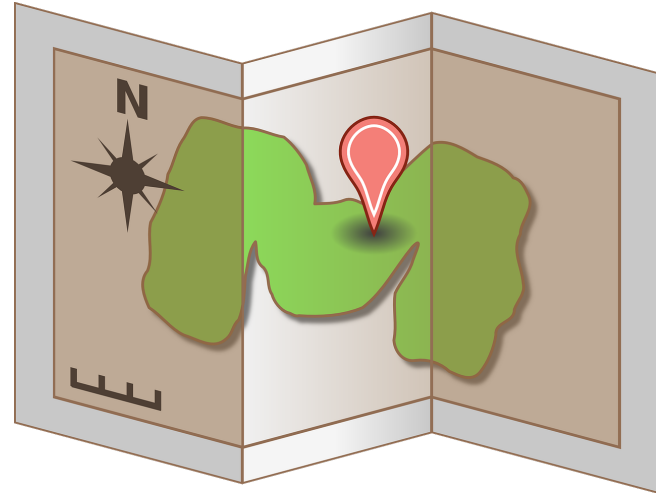


# Datetime and coordinates

# Datetime and coordinates



# Date and time

1. Periodicity
2. Time since
3. Difference between dates

# Date and time

## 1. Periodicity

Day number in week, month, season, year  
second, minute, hour.

## 2. Time since

## 3. Difference between dates

# Date and time

## 1. Periodicity

Day number in week, month, season, year  
second, minute, hour.

## 2. Time since

a. Row-independent moment

For example: since 00:00:00 UTC, 1 January 1970;

b. Row-dependent important moment

Number of days left until next holidays/ time passed  
after last holiday.

## 3. Difference between dates

# Date and time

## 1. Periodicity

Day number in week, month, season, year  
second, minute, hour.

## 2. Time since

a. Row-independent moment

For example: since 00:00:00 UTC, 1 January 1970;

b. Row-dependent important moment

Number of days left until next holidays/ time passed  
after last holiday.

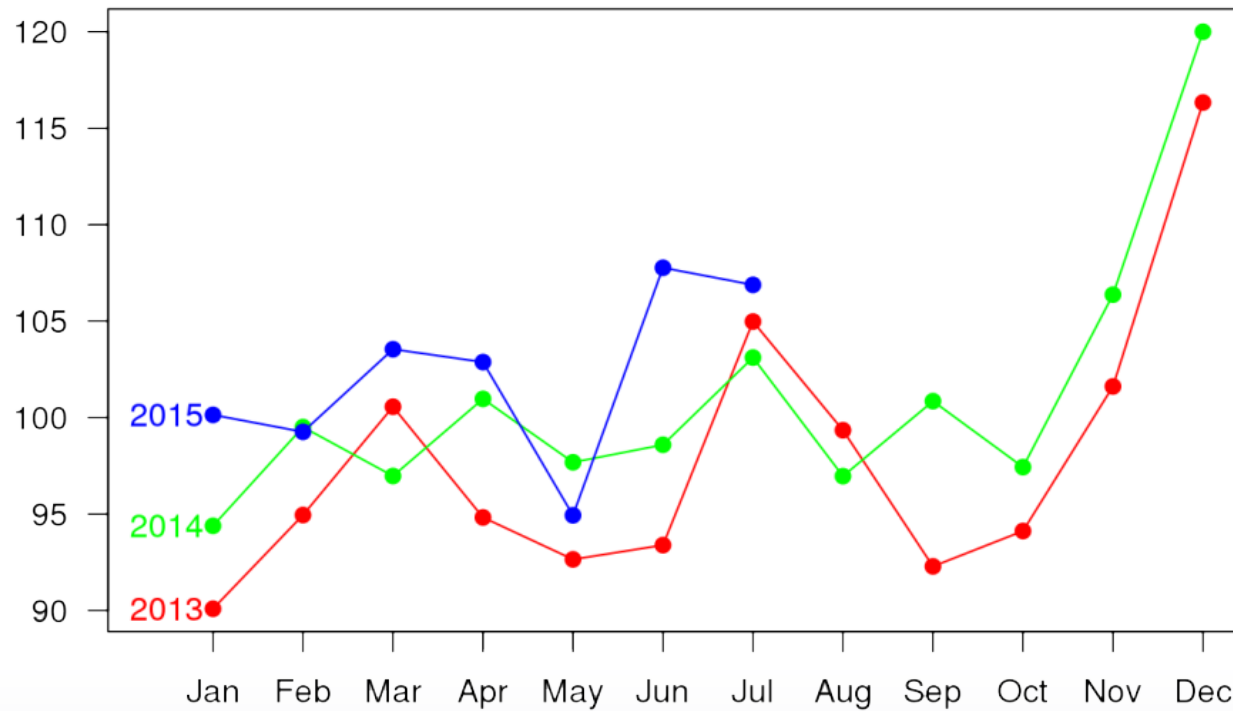
## 3. Difference between dates

`datetime_feature_1 - datetime_feature_2`

# Date and time

ROSSMANN

Seasonal plot: SalesTS



Rossmann Store Sales, <https://www.kaggle.com/thie1e/exploratory-analysis-rossmann>

# Periodicity. «Time since»

| Date       | weekday | daynumber_since_<br>year_2014 | is_holiday | days_till_h<br>olidays | <i>sales</i> |
|------------|---------|-------------------------------|------------|------------------------|--------------|
| 01.01.2014 | 5       | 0                             | True       | 0                      | 1213         |
| 02.01.2014 | 6       | 1                             | False      | 3                      | 938          |
| 03.01.2014 | 0       | 2                             | False      | 2                      | 2448         |
| 04.01.2014 | 1       | 3                             | False      | 1                      | 1744         |
| 05.01.2014 | 2       | 4                             | True       | 0                      | 1732         |
| 06.01.2014 | 3       | 5                             | False      | 9                      | 1022         |

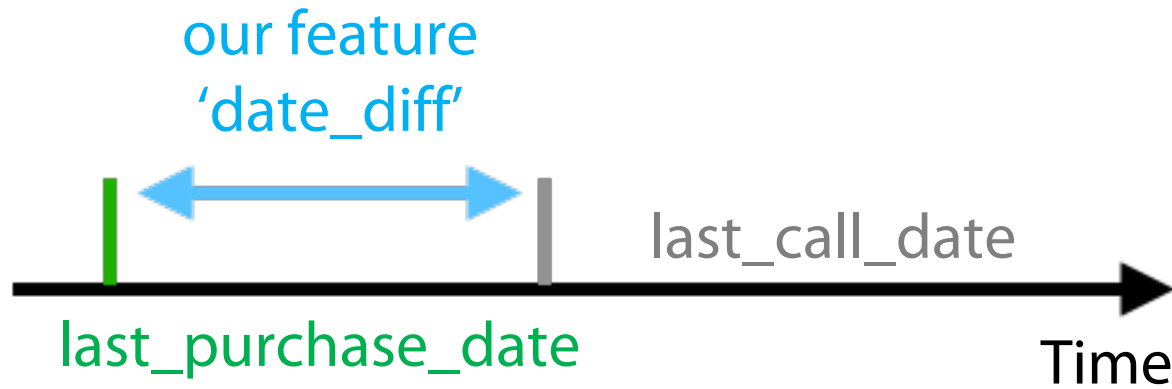
# Difference between dates

## Churn prediction

| user_id | registration_date | <i>last_purchase_date</i> | <i>last_call_date</i> | date_diff | churn |
|---------|-------------------|---------------------------|-----------------------|-----------|-------|
| 14      | 10.02.2016        | 21.04.2016                | 26.04.2016            | 5         | 0     |
| 15      | 10.02.2016        | 03.06.2016                | 01.06.2016            | -2        | 1     |
| 16      | 11.02.2016        | 11.01.2017                | 11.01.2017            | 1         | 1     |
| 20      | 12.02.2016        | 06.11.2016                | 08.02.2017            | 94        | 0     |

# Difference between dates

For a particular user:



# Competitions with coordinates

Western Australia Rental Prices

**Deloitte.**

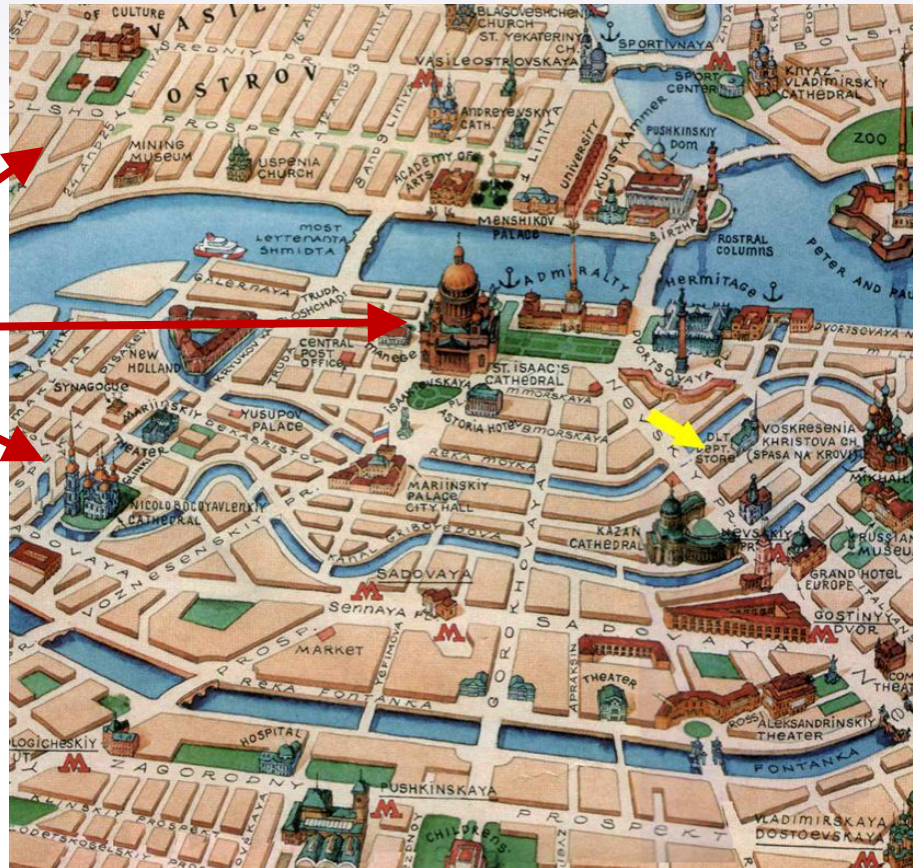


Sberbank Russian Housing Market



# Coordinates

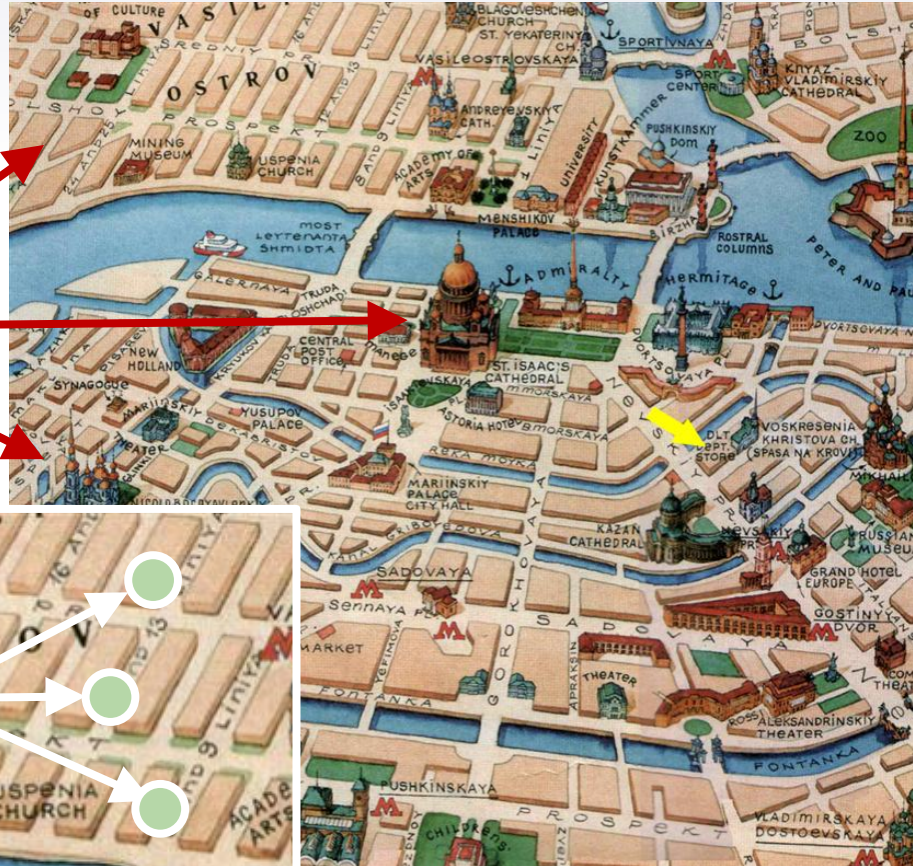
Additional  
data



St. Petersburg

# Coordinates

Additional  
data

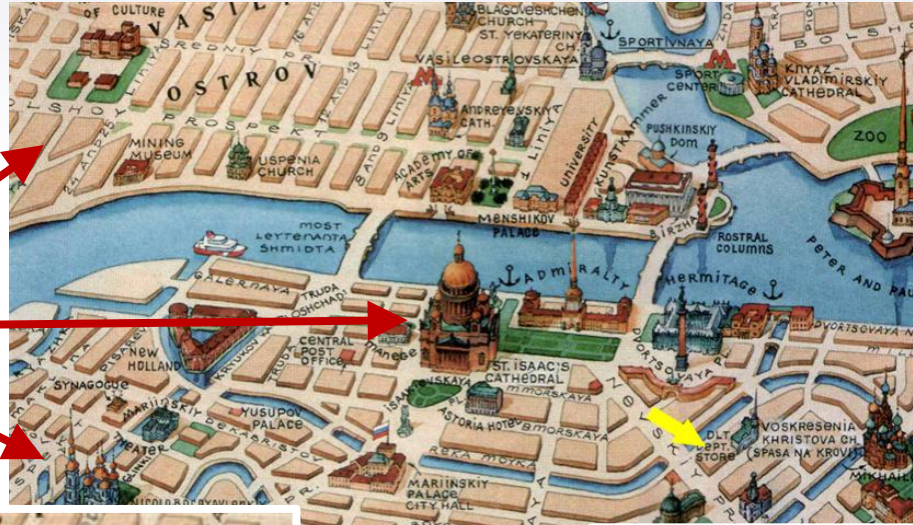


Other train samples  
and centers of  
clusters

St. Petersburg

# Coordinates

Additional  
data



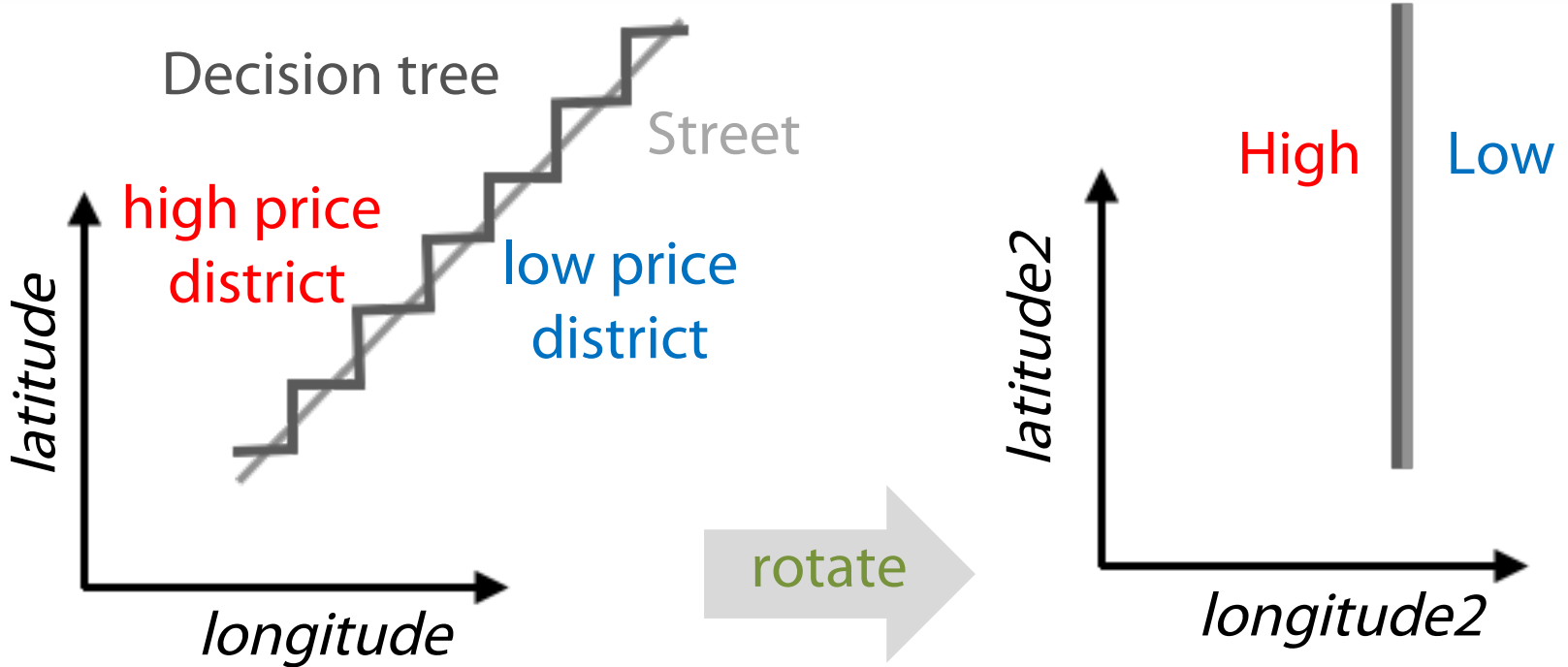
Other train samples  
and centers of  
clusters



Aggregated  
stats

St. Petersburg

# Coordinates



# Conclusion

## 1. Datetime

- a. Periodicity
- b. Time since row-independent/dependent event
- c. Difference between dates

## 2. Coordinates

- a. Interesting places from train/test data or additional data
- b. Centers of clusters
- c. Aggregated statistics